# MOBILE PHONE CHARGER USING SOLAR PANEL

## MICRO PROJECT REPORT

*Submitted by*


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# BONAFIDE CERTIFICATE

This is to certify that the micro project work entitled "**MOBILE PHONE CHARGER USING SOLAR PANEL**" is a bonafide work done by **MANOJ R M [REGISTER NO : 24UCS133]**, **THAARUN AAKASH I [REGISTER NO : 24UCS264]**, **NAVEEN PS [REGISTER NO : 24UCS150]** in partial fulfillment of the requirement, for the award of B.Tech Degree in Computer Science and Engineering by Pondicherry University during the academic year 2026-2027.

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# ABSTRACT

Solar energy is one of the most promising renewable sources of energy because it is clean, freely available and environmentally friendly. With the increasing use of mobile phones for communication, education and daily activities, the need for convenient and sustainable charging methods has also increased. Conventional mobile phone chargers depend on electrical power from the grid, which may not be available in remote and outdoor locations. This project presents a simple and portable **Solar Powered Mobile Phone Charger** that utilizes sunlight to generate electrical energy for charging a mobile phone.

In the conventional charging system, a mobile phone is connected to an adapter that obtains electrical power from the mains supply. Although this method is convenient, it is completely dependent on the availability of grid electricity. Solar-powered charging provides an alternative by directly converting sunlight into electrical energy using a photovoltaic panel. The proposed micro project is intentionally designed as a **simple batteryless system** to demonstrate the basic conversion and utilization of solar energy without introducing the additional complexity of energy storage.

The proposed system consists of a **solar panel, voltage regulation/DC-DC conversion stage and USB charging output**. When sunlight falls on the solar panel, photovoltaic cells convert the light energy into DC electrical energy. The generated electrical power is then passed through the regulation stage to obtain a suitable and stable output for the mobile charging interface. The regulated power is supplied to the mobile phone through a USB connection. Since the system does not contain a battery, charging depends directly on the availability and intensity of sunlight and the output capability of the selected solar panel and regulation circuit.

The main objective of this project is to demonstrate a simple, low-complexity and renewable method of mobile phone charging using solar energy. The system reduces dependence on conventional electrical power during suitable sunlight conditions and can be useful for basic outdoor and educational applications. The project also provides a foundation for future development. In the subsequent mini project, a rechargeable battery can be incorporated to store the generated energy and provide charging when sunlight is unavailable, along with improved power-management and monitoring features.

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# LIST OF ABBREVIATIONS

| ABBREVIATION | EXPANSION |
|---|---|
| PV | Photovoltaic |
| DC | Direct Current |
| USB | Universal Serial Bus |
| DC-DC | Direct Current to Direct Current |
| IC | Integrated Circuit |
| V | Volt |
| A | Ampere |
| W | Watt |
| GND | Ground |
| MPPT | Maximum Power Point Tracking |

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# CHAPTER – 1

# 1. INTRODUCTION

## 1.1 OVERVIEW

The increasing demand for mobile phones has created a continuous need for reliable and convenient charging facilities. Mobile phones are widely used for communication, education, information access, online services and various daily activities. Conventional mobile phone charging generally depends on electricity supplied through the electrical grid. Although this method is widely available, access to conventional electricity may be difficult in outdoor locations, remote areas and situations where a power outlet is not readily available. This creates an opportunity to explore renewable energy sources for small-scale electronic applications.

Solar energy is one of the most readily available renewable energy sources. It is obtained from sunlight and can be converted directly into electrical energy using photovoltaic (PV) technology. A solar panel consists of photovoltaic cells that produce Direct Current (DC) electrical energy when exposed to sunlight. This generated electrical energy can be conditioned using an appropriate voltage regulation or DC-DC conversion stage and supplied to a suitable electrical load.

The **Solar Powered Mobile Phone Charger** proposed in this micro project is a simple system that demonstrates the use of solar energy for mobile phone charging. The system consists of a solar panel, a suitable voltage regulation/DC-DC conversion stage and a Universal Serial Bus (USB) output interface connected to a mobile phone. The solar panel acts as the primary energy source. When sunlight falls on the panel, it converts solar energy into DC electrical energy. The generated electrical output is then passed through the regulation stage to provide a suitable output for the charging interface.

A major characteristic of this micro project is that **no battery is used for energy storage**. The project is intentionally kept simple so that the basic process of solar energy generation and direct power utilization can be understood and demonstrated. Therefore, the availability and intensity of sunlight directly influence the electrical output of the system. Under suitable sunlight conditions, the generated power can be conditioned and supplied to the mobile phone through the USB interface.

The project provides an introductory practical implementation of renewable-energy-based charging. It combines basic concepts from solar photovoltaic technology, electrical power conditioning and mobile charging into a single prototype. The project also provides a foundation for further development. In the future Mini Project, battery storage and additional

power-management features can be incorporated to overcome the limitation of direct sunlight dependence.

The overall concept of the proposed system can be represented as:

**Sunlight → Solar Panel → Voltage Regulation / DC-DC Conversion → USB Output → Mobile Phone**

Thus, the project demonstrates how solar energy can be utilized for a practical and commonly required application while keeping the design simple, economical and suitable for a first-year micro project.

## 1.2 BACKGROUND OF SOLAR ENERGY

Energy is essential for almost every modern activity, including transportation, communication, manufacturing, agriculture and electronic devices. Conventional energy generation has traditionally depended heavily on sources such as coal, petroleum and natural gas. The increasing demand for energy and concerns related to environmental impact have encouraged the development and use of renewable energy sources.

Renewable energy is obtained from naturally replenished sources. Major renewable energy sources include solar energy, wind energy, hydropower, biomass and geothermal energy. Among these sources, solar energy is particularly useful for small electronic applications because sunlight is widely available and can be directly converted into electrical energy.

Solar energy is the radiant energy received from the Sun. The energy reaching the Earth's surface can be converted into useful forms of energy using appropriate technologies. Solar photovoltaic systems are one of the important technologies used for converting sunlight directly into electricity.

The basic principle behind solar electricity generation is the **photovoltaic effect**. When sunlight falls on a photovoltaic material, photons transfer energy to electrons within the material. This process produces an electrical potential and allows current to flow through an external circuit when a suitable path is provided.

Solar energy has several characteristics that make it suitable for portable applications:

- It is a renewable source of energy.
- Sunlight is naturally available and does not require fuel transportation.
- Photovoltaic panels can directly generate DC electricity.
- Solar panels can be used in portable and small-scale systems.

- The system can operate without depending directly on the electrical grid.
- Solar energy can be used for low-power electronic applications.

However, solar energy generation is dependent on environmental conditions. The electrical output of a solar panel changes according to factors such as sunlight intensity, panel orientation, shading and environmental conditions. Therefore, a solar-powered charging system must consider the available solar power when designing the electrical interface.

In this project, solar energy is used as the primary source for mobile phone charging. Instead of storing the generated energy in a battery, the generated DC power is directly conditioned and supplied to the mobile phone through the charging interface. This makes the system simpler and allows the fundamental operation of photovoltaic energy conversion to be demonstrated clearly.

## 1.3 SOLAR PHOTOVOLTAIC TECHNOLOGY

Solar photovoltaic technology is a method of converting sunlight directly into electrical energy. The term **Photovoltaic (PV)** is derived from "photo," referring to light, and "voltaic," referring to electricity. A photovoltaic system uses semiconductor materials to convert incident sunlight into electrical power.

A solar panel consists of multiple photovoltaic cells connected together to obtain the required electrical characteristics. When sunlight falls on the photovoltaic cells, electrical charges are generated due to the photovoltaic effect. The resulting electrical output is generally Direct Current (DC).

The basic energy conversion process is:

**Solar Energy → Photovoltaic Conversion → DC Electrical Energy**

The electrical output of a solar panel is not necessarily constant. It varies depending on the intensity of sunlight and the operating conditions of the panel. Strong sunlight generally provides higher available power, whereas weak sunlight or shading can reduce the generated electrical output.

For a mobile charging application, the raw output from the solar panel may therefore need to be conditioned before being supplied to the charging interface. A suitable voltage regulation or DC-DC conversion stage can be used to obtain an appropriate electrical output.

The major components involved in the photovoltaic charging concept used in this project are:

### 1.3.1 Solar Panel

The solar panel is the primary energy-generating component. It receives sunlight and converts it into DC electrical energy.

### 1.3.2 Voltage Regulation / DC-DC Conversion Stage

The regulation or conversion stage conditions the electrical output obtained from the solar panel. Its exact operation depends on the selected hardware module and its electrical specifications.

### 1.3.3 USB Output Interface

The USB interface provides a convenient connection between the conditioned DC output and the mobile phone charging cable.

### 1.3.4 Mobile Phone

The mobile phone acts as the load of the proposed system. It receives electrical power from the USB charging interface when suitable operating conditions are available.

The simplified energy conversion chain is:

**Sunlight → PV Panel → DC Power → Regulation/Conversion → USB Output → Mobile Phone**

This arrangement forms the basic technical foundation of the proposed micro project.

## 1.4 EXISTING MOBILE CHARGING APPROACHES AND THEIR DRAWBACKS

Mobile phones are commonly charged using a wall adapter connected to an electrical outlet. In this approach, electrical power from the AC mains is converted into a suitable DC charging supply by the charger adapter. The mobile phone is then connected through an appropriate charging cable.

The conventional charging process can be represented as:
**AC Mains → Mobile Charger Adapter → USB/DC Output → Mobile Phone**

Although conventional charging is convenient, it has certain limitations.

### 1.4.1 Dependence on Grid Electricity

The conventional charging method requires access to an electrical outlet and therefore depends on the availability of grid electricity.

### 1.4.2 Limited Use in Remote Locations

In outdoor or remote locations where electrical outlets are unavailable, conventional chargers may not provide a convenient charging solution.

### 1.4.3 Dependence on External Power Infrastructure

Conventional chargers require an existing electrical infrastructure. This makes them less suitable for situations where access to the grid is temporarily unavailable.

### 1.4.4 Requirement for a Separate Charging Adapter

A conventional mobile phone normally requires a compatible charging adapter and cable. The charging system therefore depends on additional equipment.

### 1.4.5 Opportunity for Renewable Charging

The dependence on conventional electrical power provides an opportunity to investigate renewable energy-based charging methods. Solar energy can provide an alternative source of electrical power when sufficient sunlight is available.

The proposed project addresses this opportunity by using a solar panel as the primary power source rather than directly depending on an AC electrical outlet.

## 1.5 PROBLEM STATEMENT

Mobile phones require regular charging, but conventional charging systems depend on the availability of electrical power and suitable charging outlets. In outdoor and remote environments, access to conventional electrical power may not always be convenient.

Solar energy provides a renewable source that can be converted into electrical energy using photovoltaic technology. However, directly connecting a solar panel to a mobile phone is not an appropriate approach because the electrical output of a solar panel varies with sunlight conditions.

Therefore, the problem addressed by this project is:
To develop a simple batteryless Solar Powered Mobile Phone Charger that converts sunlight into electrical energy through a solar panel, conditions the generated DC power using a suitable regulation/conversion stage, and provides the conditioned output through a USB interface for mobile phone charging.

The project focuses on demonstrating the basic working principle without incorporating a battery-based energy-storage system.

## 1.6 OBJECTIVES OF THE PROPOSED SYSTEM

The main objective of the project is to develop a simple and practical solar-powered charging system for a mobile phone.

The specific objectives are:

1. To utilize **solar energy** as the primary source of electrical power.
2. To convert sunlight into DC electrical energy using a **Photovoltaic (PV) solar panel**.
3. To condition the generated electrical output using a suitable **voltage regulation/DCDC conversion stage**.
4. To provide the conditioned output through a **USB charging interface**.
5. To demonstrate direct solar-powered mobile phone charging **without using a battery**.
6. To study the effect of different sunlight conditions on the generated electrical output.
7. To develop a simple, low-complexity prototype suitable for a first-year micro project.
8. To provide a foundation for incorporating battery storage and improved power management in a future Mini Project.

## 1.7 SCOPE OF THE PROJECT

The scope of this project is limited to the design and development of a basic batteryless solarpowered mobile phone charging prototype.

The project includes:

- Solar energy collection using a photovoltaic panel.
- Conversion of sunlight into DC electrical energy.
- Voltage regulation or DC-DC conversion of the generated output.
- Delivery of conditioned power through a USB interface.
- Direct connection to a mobile phone as the charging load.

- Measurement and observation of electrical parameters during testing.
- Evaluation of charging behaviour under different sunlight conditions.

The project does **not** include:

- Battery-based energy storage.
- Battery charging or battery management.
- Maximum Power Point Tracking (MPPT) implementation.
- IoT-based monitoring.
- Wireless mobile charging.
- Cloud-based monitoring.
- Advanced automatic power-management systems.

## 1.8 SIGNIFICANCE OF THE PROJECT

The project is significant because it demonstrates the practical application of renewable energy to a commonly used electronic device. Instead of depending completely on conventional electrical power, the proposed system explores the use of sunlight as a direct energy source for mobile charging.

From an educational perspective, the project provides practical understanding of:

- Renewable energy utilization.
- Photovoltaic energy conversion.
- DC electrical power generation.
- Voltage regulation and power conditioning.
- USB power delivery.
- Basic electrical measurement.
- Prototype development and testing.

The project also demonstrates the importance of selecting appropriate electrical components when integrating a renewable energy source with an electronic load. Since the output of a solar panel varies with environmental conditions, suitable power conditioning is necessary before the generated energy is supplied to the mobile charging interface.

The simple architecture also makes the project suitable for demonstration and further development. The concept can be extended in a future version by adding energy storage, improved regulation and monitoring features.

## 1.9 ORGANIZATION OF THE PROJECT REPORT

This project report is organized into chapters to present the development of the Solar Powered Mobile Phone Charger in a systematic manner.

**Chapter 1 – Introduction** explains the background of solar energy, photovoltaic technology, conventional mobile charging methods, the problem statement, objectives, scope and significance of the proposed project.

**Chapter 2 – Literature Survey** presents relevant research studies related to solar energy harvesting, photovoltaic charging systems and solar-powered mobile charging. It also discusses the comparison of existing research and identifies the gap addressed by the proposed micro project.

**Chapter 3 – System Study** describes the existing mobile charging system, its limitations, the proposed batteryless solar charging system, feasibility and system requirements.

**Chapter 4 – System Requirements** presents the hardware, software/documentation and other requirements needed to develop and test the prototype.

**Chapter 5 – Proposed System** explains the architecture, workflow and individual modules of the proposed system, including the solar energy generation, voltage regulation/DC-DC conversion, USB output and mobile charging stages.

**Chapter 6 – Implementation and Results** describes the hardware connections, prototype construction, testing procedure and implementation process.

**Chapter 7 – Conclusion and Future Scope** summarizes the work completed, discusses the significance and limitations of the prototype, and presents possible future enhancements such as battery-based energy storage.

**References** contains the research papers and other academic sources referred to during the preparation of the project report.

**Appendices** provide supporting information such as the bill of materials, component specifications and prototype photographs.

# CHAPTER – 2

# 2. LITERATURE SURVEY

## 2.1 OVERVIEW

The literature survey provides an understanding of previous research and developments related to solar-powered mobile charging systems. Solar energy has been widely investigated as an alternative energy source for portable electronic devices because photovoltaic (PV) panels can convert sunlight directly into electrical energy. Researchers have developed different types of solar charging systems, ranging from simple portable chargers to larger charging stations with batteries, charge controllers, converters, wireless charging and monitoring systems.

A review of existing research is important for identifying the commonly used components, system architectures, charging methods and limitations of existing solar-powered charging solutions. It also helps in selecting an appropriate approach for the development of the proposed micro project.

Several studies have investigated solar-powered charging for mobile phones and other portable devices. Some systems use batteries to store solar energy and provide charging when sunlight is unavailable, while others focus on portable direct-use charging, charging stations or wireless charging. For example, Manchester and Swan experimentally studied the power and energy requirements of off-grid mobile phone charging and examined the sizing of solar panels, batteries and inverters for such applications.

Other research has focused on practical solar charging infrastructure. Umarani et al.

developed a solar docking station for smartphones and laptops using photovoltaic panels, batteries, charge-control circuitry and power-conversion stages, demonstrating a larger standalone charging arrangement for outdoor use.

The present project takes a deliberately simpler approach. Instead of developing a batterybased charging station or an advanced wireless charging system, the proposed micro project focuses on demonstrating the basic energy path:

Sunlight → Solar Panel → Voltage Regulation / DC-DC Conversion → USB Output → Mobile Phone

The literature reviewed in this chapter therefore provides the technical background for selecting a simple architecture while identifying the additional features that can be considered in future development.

## 2.2 LITERATURE SURVEY

### 2.2.1 Off-Grid Mobile Phone Charging – Manchester and Swan

Manchester and Swan studied the requirements associated with off-grid mobile phone charging. Their work examined the power and energy requirements of mobile phone charging and investigated the efficiency of low-power inverters. The study also considered the sizing of batteries, inverters and solar panels for mobile charging applications.

The study demonstrates that mobile phone charging can be an important electrical requirement in areas where conventional grid electricity is unavailable or limited. It also highlights the importance of understanding the electrical power requirements of the mobile charging load before designing a renewable-energy-based charging system.

Relevance to the Proposed Project

The study provides useful background for understanding the relationship between:

- Solar generation
- Available electrical power
- Charging load
- Power-conversion requirements

Our project applies the same general concept of using solar energy for mobile charging but uses a much simpler batteryless architecture suitable for a first-year micro project.

### 2.2.2 Design and Implementation of a Solar Docking Station – Umarani et al.

Umarani et al. presented a solar docking station for smartphones and laptops intended for outdoor and open-space use. The study was motivated by situations where access to conventional electrical outlets is difficult. The system uses photovoltaic panels together with batteries, charge-control circuitry and power-conversion stages. It also provides USB ports for direct DC charging of mobile devices.

The work demonstrates that solar energy can be integrated into a stand-alone charging infrastructure. The inclusion of USB outputs is particularly relevant to mobile-device Charging because USB provides a practical interface between the power-conditioning system and consumer electronics.

Relevance to the Proposed Project

The study supports the use of a solar panel and USB-based charging interface in our project. However, its architecture is considerably more complex because it includes energy storage and larger charging infrastructure.

Our micro project reduces the architecture to the essential stages required for demonstrating direct solar-powered mobile charging.

### 2.2.3 Photovoltaic Mobile Charging System Design – Başoğlu et al.

Başoğlu et al. developed a photovoltaic mobile charging system in the form of a solar pole and implemented a prototype at Gümüşhane University. The study describes a photovoltaicbased outdoor charging arrangement and reports mobile charging through a 5 V, 2 A output.

The system was designed for use in open-air conditions and demonstrates the practical application of solar energy for mobile charging infrastructure.

The research demonstrates that photovoltaic charging can be implemented as a practical infrastructure for public or campus environments. It also emphasizes the importance of obtaining a suitable regulated output for mobile devices.

**Relevance to the Proposed Project**

This study supports the importance of regulated DC output for mobile charging. Our project follows the same general principle but reduces the system size and complexity for a small prototype.

The solar pole concept is intended for a larger outdoor installation, whereas our project is intended as a simple portable/educational prototype.

### 2.2.4 Design of Solar-Powered Charging Backpack – Taverne et al.

Taverne et al. investigated the design of a solar-powered charging backpack capable of charging a mobile phone. The research involved reviewing existing products, comparing solar-cell technologies and regulators, designing the charging system and testing the completed system before integrating it into the backpack. The study reported successful mobile phone charging.

This research demonstrates the importance of selecting an appropriate solar-cell technology and regulator when designing a portable charging system. It also shows that solar charging can be incorporated into portable everyday-use products.

Relevance to the Proposed Project

The research is relevant to our project because both systems focus on portable solar-powered mobile charging. However, our micro project is much simpler and focuses on demonstrating the fundamental photovoltaic-to-mobile charging process rather than integrating the charger into a wearable product.

### 2.2.5 Design and Development of a Portable Solar-Powered Mobile Charging System – Goudar

Goudar presented a portable solar-powered mobile charging system using photovoltaic technology. The reported design included a solar panel, lithium-ion battery, TP4056 charging

module, voltage regulator and boost converter to obtain a suitable output for mobile charging. Experimental validation was performed under sunlight conditions.

The study demonstrates a more complete portable solar charging architecture in which energy is first stored in a battery and subsequently delivered to the mobile phone through regulated power-conversion stages.

Relevance to the Proposed Project

This study is particularly useful for understanding how a battery-based solar charger can be constructed. However, the present micro project intentionally does not include a battery or battery-charging module.

The battery-based architecture can therefore be considered as a possible direction for our future Mini Project, while the current Micro Project concentrates on the simpler direct solarpower path.

### 2.2.6 Portable Solar-Powered Wireless Charger

Recent research has also investigated wireless charging using solar energy. Yuwono et al. presented a portable solar-powered wireless charger using a photovoltaic panel and near-field inductive wireless power transfer. Their work focused on designing and experimentally evaluating a portable wireless charging system.

Wireless charging provides convenience by removing the need for a direct charging cable. However, it requires additional power-transfer circuitry and alignment considerations compared with a basic wired USB charging interface.

Relevance to the Proposed Project

The study demonstrates a more advanced direction for solar-powered mobile charging. Our project deliberately avoids wireless charging because the objective of the micro project is to understand the basic solar energy generation and regulated wired power-delivery process.

### 2.2.7 Portable Solar-Powered Wireless Charging Device – Kok et al.

Kok et al. developed a portable solar-powered wireless charging device with an advanced charging algorithm. The reported system included a solar panel and a 6500 mAh lithium-ion battery and was designed to charge smartphones and laptops.

The research shows how solar charging can be combined with energy storage and advanced charging control to provide a more flexible charging solution.

Relevance to the Proposed Project

The study is useful for understanding possible future improvements such as:

- Battery energy storage
- Advanced charging control
- Wireless charging
- Portable multi-device charging

However, these features are outside the scope of our present first-year Micro Project.

### 2.2.8 Solar-Powered Multi-Functional Charging Systems

Once such system integrates a solar panel, charge controller, sensors and an IoT module to provide real-time monitoring and multiple charging applications.

Such systems demonstrate how a basic solar charging concept can be expanded into a smart energy-management platform.

Relevance to the Proposed Project

Although monitoring and IoT features can improve the functionality of a charging system, they are not necessary for demonstrating the basic working principle of the proposed micro project.

Therefore, the present project intentionally avoids unnecessary complexity and concentrates on the core energy-conversion process.

## 2.3 COMPARATIVE ANALYSIS OF EXISTING RESEARCH

The reviewed studies demonstrate different approaches to solar-powered charging. Some focus on direct portable charging, while others incorporate batteries, wireless charging, charging stations or monitoring systems.

**Table 2.1 Comparative Analysis of Existing Research**

The comparison shows that many existing systems introduce energy storage, advanced conversion, wireless power transfer or monitoring capabilities. These features are useful for practical large-scale systems but increase the complexity of the design.

The proposed micro project is intentionally different in scope. It focuses on the fundamental operation of converting sunlight into electrical energy and supplying the conditioned electrical output to a mobile phone through a USB interface.

| STUDY | MAIN APPROACH | BATTERY | CHARGING METHOD | MAJOR FEATURE |
|---|---|---|---|---|
| Manchester & Swan | Off-grid mobile charging | Yes / considered | Wired | Power and energy analysis |
| Umarani et al. | Solar docking station | Yes | USB + AC | Outdoor charging station |
| Başoğlu et al. | PV solar pole | System-level storage/design | Wired | Campus/openair charging |

| | | | | |
|---|---|---|---|---|
| Taverne et al. | Solar charging backpack | Designdependent | Wired | Portable charging |
| Goudar | Portable solar charger | Yes | Wired | Battery + regulator + boost |
| Yuwono et al. | Solar wireless charger | Systemdependent | Wireless | Inductive charging |
| Kok et al. | Portable solar wireless charger | Yes | Wireless | Advanced charging |
| Proposed Micro Project | Simple solar mobile charger | No | USB wired | Batteryless direct solar charging |

## 2.4 KEY GAPS IDENTIFIED

The literature survey identifies several common characteristics of existing solar-powered charging systems.

### 2.4.1 High System Complexity

Many existing systems combine multiple components such as batteries, charge controllers, boost converters, inverters, wireless charging circuits and monitoring systems.

For a first-year educational project, implementing all these features would increase complexity unnecessarily.

### 2.4.2 Dependence on Energy Storage

Several portable solar charging systems use batteries to store generated energy. This improves charging availability when sunlight is unavailable but adds additional components and battery-management requirements.

The present micro project addresses this by intentionally operating without battery storage.

### 2.4.3 Advanced Charging Technologies

Some recent systems use wireless charging, intelligent charging algorithms or IoT-based monitoring.

Although these technologies are valuable for advanced systems, they are beyond the objective of the present micro project.

### 2.4.4 Need for a Simple Educational Prototype

The reviewed literature provides many advanced implementations, but a simple system that demonstrates the basic sequence of:

Solar Energy → PV Conversion → Voltage Conditioning → USB Output → Mobile Phone

is more appropriate for an introductory micro project.

### 2.4.5 Dependence on Sunlight

A batteryless system has an important limitation: the available charging power depends directly on the solar energy available at the time of operation. This makes sunlight conditions an important factor in system performance.

### 2.4.6 Identified Project Gap

Based on the literature reviewed, the proposed project focuses on a low-complexity, batteryless solar mobile phone charger that demonstrates the fundamental photovoltaic charging concept without incorporating unnecessary advanced features.

The identified gap can therefore be stated as:

There is a need for a simple educational prototype that demonstrates direct solar-powered mobile charging using a photovoltaic source and regulated DC output while avoiding the additional complexity of battery storage, wireless charging and smart monitoring.

## 2.5 SUMMARY

This chapter reviewed research and development related to solar-powered mobile phone charging systems. The studies examined different approaches, including off-grid charging, solar docking stations, photovoltaic charging poles, portable charging backpacks, batterybased solar chargers and wireless charging systems.

The literature shows that solar energy can be effectively utilized for mobile charging, but many existing systems introduce additional components such as batteries, charge controllers, boost converters, wireless power-transfer circuits and monitoring technologies.

Based on the findings of the literature survey, the present project adopts a simpler architecture suitable for a first-year micro project. The proposed system focuses on converting sunlight into DC electrical energy using a solar panel, conditioning the generated power through a suitable voltage regulation/DC-DC conversion stage and delivering the output through a USB interface to the mobile phone.

The project intentionally excludes battery storage from the current implementation. Batterybased storage, improved power management and other advanced features can be considered in the subsequent Mini Project.

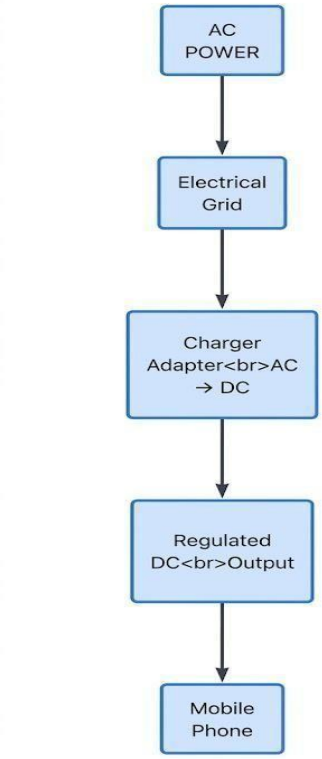
Thus, the literature survey establishes the technical background and justification for developing a simple, batteryless Solar Powered Mobile Phone Charger.

# CHAPTER – 3

# 3. SYSTEM STUDY

## 3.1 EXISTING SYSTEM

Conventional mobile phones are charged using a charger connected to the electrical grid. Grid Supply → Charger Adapter → Regulated DC → Mobile Phone



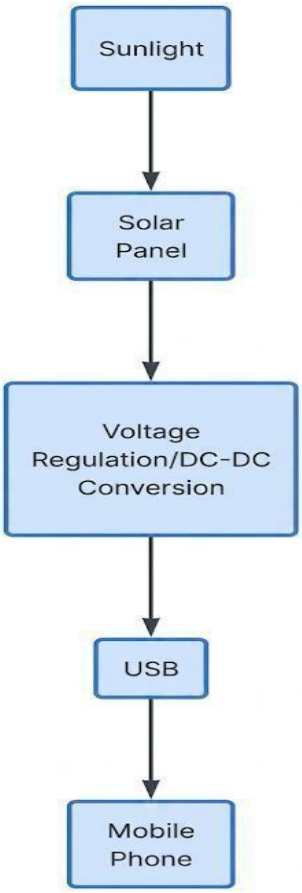

## 3.2 LIMITATIONS OF EXISTING SYSTEM

- Depends on grid electricity.
- Requires an electrical outlet.
- Not convenient in remote/outdoor locations.
- Does not directly utilize solar energy.

### 3.3 PROPOSED SYSTEM

The proposed system uses a solar panel to convert sunlight into electrical energy. The output is regulated and supplied through a USB port to charge the mobile phone.

Sunlight → Solar Panel → Voltage Regulation/DC-DC Conversion → USB → Mobile Phone




Main features:

- Uses renewable solar energy.
- Simple and portable.
- Batteryless operation.
-

### 3.4 FEASIBILITY STUDY

| TYPE | DESCRIPTION |
|---|---|
| Technical | Uses simple and available components. |
| Economic | Low-cost prototype. |
| Operational | Easy to operate. |
| Environmental | Uses renewable solar energy. |

### 3.5 FUNCTIONAL REQUIREMENTS

- Convert sunlight into electrical energy.
- Provide suitable DC output.
- Regulate/condition the solar output.
- Provide USB output.
- Charge a mobile phone under suitable sunlight conditions.

### 3.6 NON-FUNCTIONAL REQUIREMENTS

- Simple: Easy to understand and operate.
- Portable: Can be used outdoors.
-

# CHAPTER – 4

# 4. SYSTEM REQUIREMENTS

## 4.1 HARDWARE REQUIREMENTS

The hardware required for the Solar Powered Mobile Phone Charger is:

- Solar panel
- Voltage regulation / DC-DC conversion module
- USB output connector
- Connecting wires
- Mobile phone for testing
- Supporting frame or base

## 4.2 SOFTWARE REQUIREMENTS

The proposed system is primarily a hardware-based project and does not require any programming software.

- Operating System: Windows/Linux (for documentation only)
- Documentation: MS Word
- No microcontroller programming is required.
-

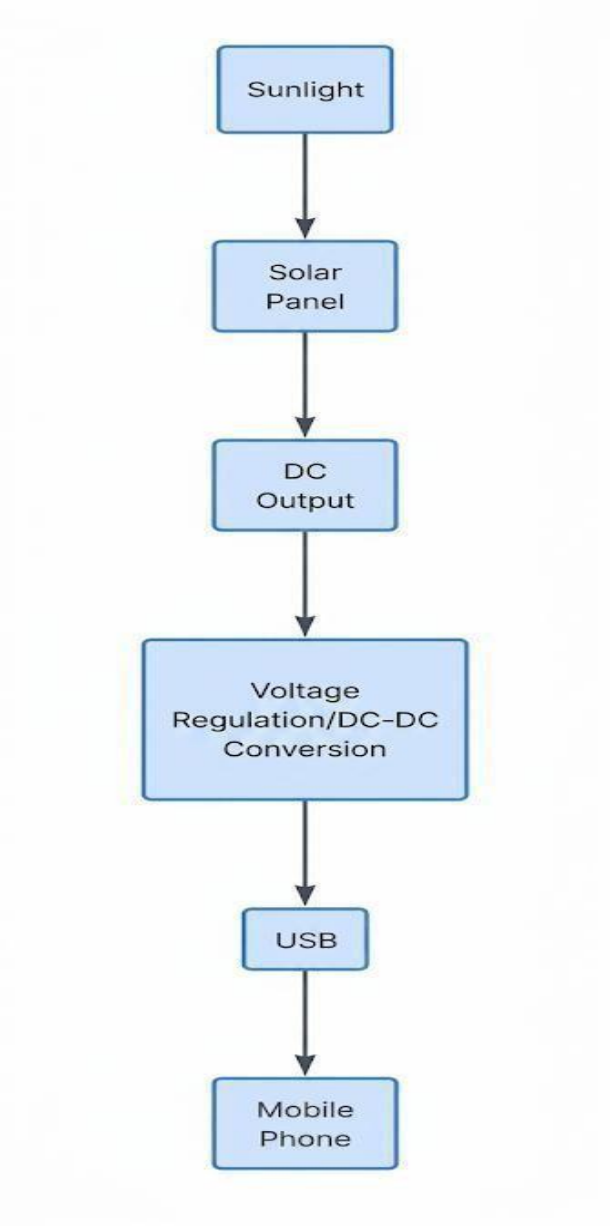
## 4.3 COMPONENT SPECIFICATIONS

| COMPONENT | PURPOSE |
|---|---|
| Solar Panel | Converts sunlight into DC electrical energy |
| Voltage Regulation / DC-DC Module | Conditions the solar panel output |
| USB Connector | Provides output connection |
| Connecting Wires | Provides electrical connections |
| Mobile Phone | Acts as the charging load |

## 4.4 POWER REQUIREMENTS

The system obtains its power directly from the solar panel.

Sunlight → Solar Panel → DC Output → Voltage Regulation/DC-DC → USB → Mobile Phone




- No battery storage is used in the current Micro Project.
- Output depends on the available sunlight.
- Solar panel orientation and sunlight intensity affect the output.
- The output must be verified before connecting the mobile phone.

## 4.5 ENVIRONMENTAL REQUIREMENTS

The system should be operated under suitable environmental conditions.

- Adequate sunlight should be available.
- The solar panel should be properly exposed to sunlight.
- Avoid water and excessive moisture.
- Avoid physical damage to the panel and electrical connections.
- Very low sunlight may reduce the available output.
- Proper ventilation should be maintained around electronic components.

# CHAPTER – 5

# 5. PROPOSED SYSTEM

## 5.1 SYSTEM ARCHITECTURE

The proposed system is designed to convert solar energy into a suitable electrical output for charging a mobile phone.

The system follows a simple batteryless architecture consisting of a solar energy generation stage, voltage regulation/DC-DC conversion stage, USB output stage, and mobile phone charging stage.

The solar panel acts as the energy source, while the regulation/DC-DC stage conditions the generated output before it reaches the mobile phone.




## 5.2 SYSTEM WORKFLOW

The system operates through a sequence of energy conversion and power delivery stages.

**Working Flow**

1. Sunlight falls on the solar panel.
2. The photovoltaic panel converts sunlight into DC electrical energy.
3. The generated output is supplied to the voltage regulation/DC-DC conversion module.
4. The module conditions the electrical output to a suitable level.
5. The conditioned output is provided through the USB interface.

6. The mobile phone is connected to the USB output.

7. When sufficient solar power is available, the mobile phone receives charging power.

8. The charging performance varies according to the available sunlight.

**Figure 5.2 – Energy Flow Diagram**

Solar Energy

↓

PV Conversion

↓

DC Electrical Energy ↓

Voltage Conditioning

↓

USB Power Output

↓

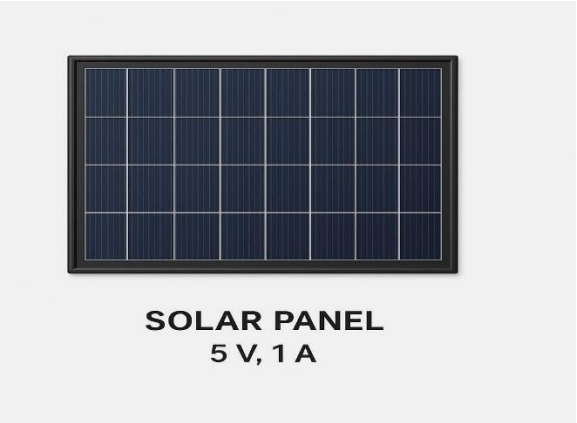
Mobile Phone

## 5.3 SOLAR ENERGY GENERATION MODULE

The solar energy generation module is the first stage of the proposed system. It consists of a Photovoltaic (PV) solar panel.

The solar panel converts solar radiation into electrical energy through the photovoltaic effect.

The generated electrical energy is in DC form and is supplied to the next stage of the system.




The output of the solar panel depends on factors such as:

- Intensity of sunlight
- Panel orientation
- Weather conditions
- Panel characteristics
- Time of the day

Since the current prototype does not include battery storage, the generated energy is directly passed to the voltage conditioning stage.

## 5.4 VOLTAGE REGULATION / DC-DC CONVERSION MODULE

The voltage regulation/DC-DC conversion module is used to condition the electrical output obtained from the solar panel.

The solar panel output can vary depending on the available sunlight. Therefore, the output must be appropriately conditioned before being supplied to the USB interface.




The main functions of this stage are:

- Receive DC power from the solar panel.
- Condition the incoming voltage.
- Provide a suitable output for the USB stage.
- Improve the suitability of the generated power for the connected load.

The exact module type and electrical ratings should be specified only after verifying the actual module used in the prototype.

## 5.5 USB OUTPUT MODULE

The USB output module provides the interface between the voltage conditioning stage and the mobile phone.

The conditioned DC output is connected to the USB interface. A suitable USB cable can then be used to connect the output to the mobile phone.

The USB stage provides:

- Convenient charging connection.
- Standard physical interface.

- Connection between the power-conditioning stage and mobile phone.

The output voltage should be verified before connecting the mobile phone.

## 5.6 MOBILE PHONE CHARGING MODULE

The mobile phone acts as the final load of the proposed system.

When sufficient solar energy is available, the conditioned electrical output is supplied to the phone through the USB connection. The phone uses this electrical energy for charging its internal battery.

The charging performance depends on:

- Available sunlight.
- Solar panel output.
- Voltage conditioning.
- Power requirement of the phone.
- Electrical losses in the system.

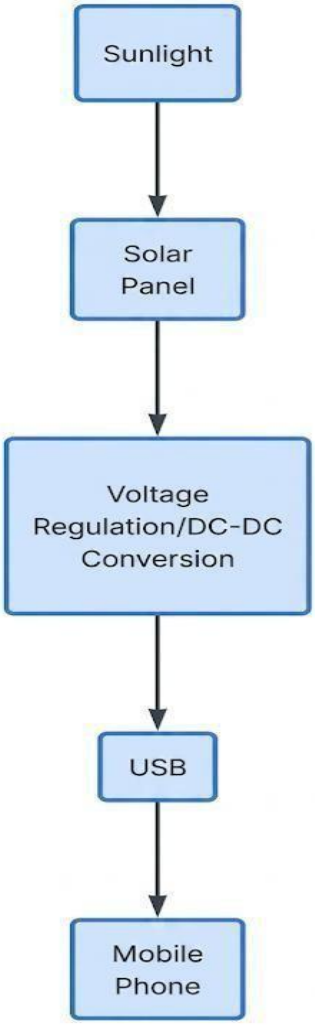
As the current project is batteryless, charging may reduce or stop when the available solar power becomes insufficient.

## 5.7 OVERALL WORKING OF THE PROPOSED SYSTEM

The complete system begins with sunlight falling on the solar panel. The Photovoltaic (PV) panel converts the solar energy into DC electrical energy. This output is then passed to the voltage regulation/DC-DC conversion stage, where it is conditioned for the required output.

The conditioned power is supplied through the USB output interface. When a mobile phone is connected and sufficient solar power is available, the phone receives electrical power for charging.

The complete operation can therefore be represented as:



# CHAPTER-6

# 6.IMPLEMENTATION AND RESULT

## 6.1 SYSTEM IMPLEMENTATION

The proposed system was implemented using a solar panel, voltage regulation/DC-DC conversion stage, USB output, and a mobile phone as the load.

THE IMPLEMENTATION FOLLOWS THE SEQUENCE:

Sunlight → Solar Panel → Voltage Regulation/DC-DC Conversion → USB Output → Mobile Phone

The solar panel generates DC electrical energy from sunlight. The generated output is connected to the voltage conditioning stage, which provides a suitable output for the USB interface. The USB output is then connected to the mobile phone for charging.

The system is designed without battery storage, making the prototype simple and suitable for demonstrating direct solar-powered charging.

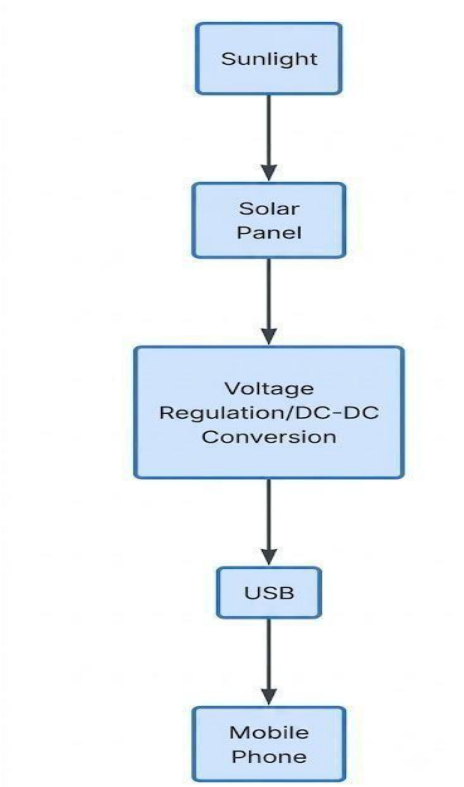
## 6.2 HARDWARE CONNECTIONS

The hardware components are connected in the following order:

1. The solar panel is connected to the input of the voltage regulation/DC-DC conversion stage.
2. The output of the conditioning stage is connected to the USB output.
3. A USB cable is connected between the USB output and the mobile phone.
4. All electrical connections are checked before operation.

5. The solar panel is exposed to suitable sunlight during testing.

**Figure 6.1 – Functional Connection Diagram**




Note: The exact module name and ratings should be updated after verifying the actual module used in your prototype.

## 6.3 PROTOTYPE CONSTRUCTION

The prototype was constructed by arranging the solar panel and electrical components in a suitable configuration. The components were connected using appropriate wires and supported securely for demonstration.

The solar panel was positioned so that it could receive sunlight. The voltage regulation/DCDC conversion stage was connected between the solar panel and USB output.

**Figure 6.2 – Working Flowchart of the Solar Mobile Charger**

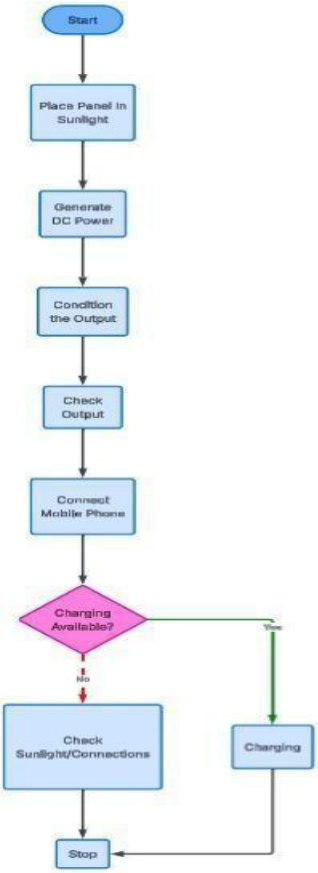



**Figure 6.3 – Prototype of the Solar Powered Mobile Phone Charger**



## 6.4 TESTING PROCEDURE

The prototype was tested to verify whether solar energy could be converted into a suitable electrical output for mobile charging.

The testing procedure consists of the following steps:

1. Place the solar panel under available sunlight.
2. Connect the solar panel to the voltage regulation/DC-DC conversion stage.
3. Measure or verify the output of the conditioning stage.
4. Connect the USB output to the mobile phone.
5. Observe the charging response.
6. Record the output under different sunlight conditions.
7. Repeat the test when required for validation.

**Figure 6.4 – Testing and Operating Sequence**

Solar Panel

↓

Voltage Output Check

↓

Output Conditioning

↓

USB Connection

↓

Mobile Phone Connection

↓

Observe Charging

↓

Record Results

## 6.5 TESTING AND VALIDATION

Testing and validation were performed to determine whether the prototype operates according to the intended design.

The following aspects can be checked:

- Solar panel output.
- Conditioned DC output.
- USB output.
- Mobile phone charging response.
- Effect of sunlight conditions.
- Stability of electrical connections.

The prototype should be considered successful when the system produces a suitable output under adequate sunlight and the connected mobile phone responds to the charging input.

Important: Actual measured values should be entered here after testing the physical prototype.

## 6.6 EXPERIMENTAL OBSERVATIONS

The output of the solar-powered charger can vary according to sunlight conditions. Therefore, observations should be recorded at different times or under different sunlight levels.

**Table 6.1 – Experimental Observation of Solar Panel Output**

| S. No. | Sunlight Condition | Output Voltage | Output Current |
|---|---|---|---|
| 1 | Strong sunlight | ___ V | ___ A |
| 2 | Moderate sunlight | ___ V | ___ A |
| 3 | Low sunlight | ___ V | ___ A |
| 4 | Cloudy condition | ___ V | ___ A |

## 6.7 RESULTS AND DISCUSSION

The experimental testing is expected to demonstrate that solar energy can be used as a direct power source for mobile phone charging when sufficient sunlight is available.

The observations should be analyzed based on:

- Solar panel output.
- Variation in sunlight intensity.

- Output of the voltage conditioning stage.
- USB output.
- Mobile phone charging response.

The system's performance is directly dependent on the available solar energy. Strong sunlight generally provides better conditions for power generation, while reduced sunlight can result in lower available output.

The results obtained from the actual prototype will be used to determine the effectiveness of the proposed batteryless charging system.

## 6.8 LIMITATIONS OF THE PROTOTYPE

The current prototype has certain limitations because it is designed as a simple first-year Micro Project.

1. Dependence on Sunlight: The system requires sufficient sunlight to generate useful electrical power.
2. No Battery Storage: The present prototype does not store solar energy for later use.
3. Variable Output: Solar panel output can vary with weather, time of day, and panel orientation.
4. Limited Charging Consistency: Charging may not remain continuous when available solar power becomes insufficient.
5. Basic Power Management: The current prototype does not implement advanced power-management techniques.
6. Prototype-Level Design: The system is intended primarily for educational demonstration and can be further improved in a future Mini Project.

# CHAPTER-7

# 7.CONCLUSION AND FUTURE SCOPE

## 7.1 CONCLUSION

The Solar Powered Mobile Phone Charger demonstrates the practical use of solar energy for mobile phone charging.It converts into DC electrical energy and a Voltage Regulation/DCDC Conversion Stage to condition the output before supplying it through USB.

The prototype is designed as a simple batteryless system, making it suitable for a first-year Micro Project. The study shows that mobile charging can be supported using renewable solar energy when sufficient sunlight is available.

The project also provides practical understanding of solar energy conversion, voltage conditioning, electrical connections, and prototype testing.

## 7.2 FUTURE SCOPE

The proposed system can be improved in the future to provide better performance and more convenient operation.

Possible improvements include:

- Adding battery storage to store excess solar energy.
- Improving voltage and power regulation.
- Using more efficient solar panels.
- Adding monitoring of voltage and current.

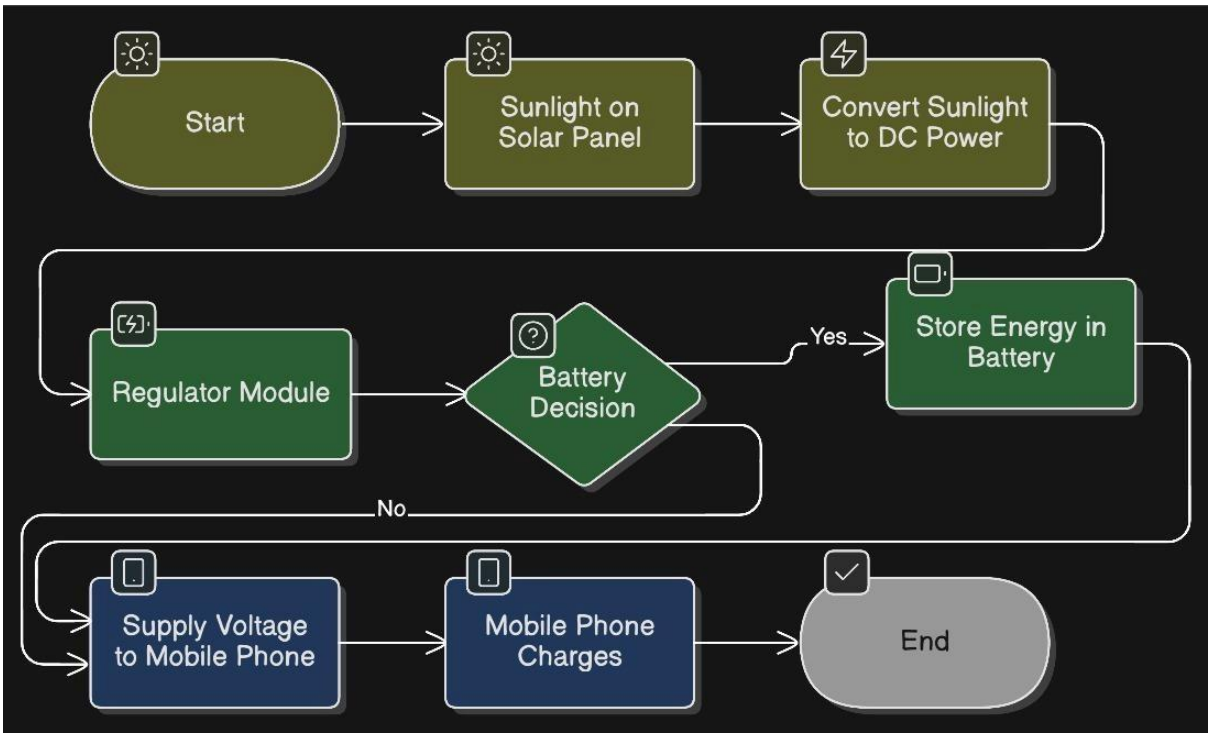
- Improving the mechanical design for portability.
- Developing better protection against over-voltage and over-current.
- Studying advanced solar power-management techniques.

These improvements can make the system more suitable for practical outdoor applications.

### 7.3 MINI PROJECT ENHANCEMENT

The current Micro Project can be extended into a more advanced Mini Project by introducing energy storage and additional power-management features.

A possible future architecture is:




The Mini Project can focus on storing solar energy during periods of sufficient sunlight and using the stored energy when sunlight is temporarily unavailable.

Additional features such as battery monitoring, improved power management, and performance monitoring can also be considered.

# REFERENCE

For your report, keep the References section limited to the four literature papers actually used in Chapter 2. Use an academic reference format and do not put website links, matching your college template.


1. Manchester, A. and Swan, L. G., "Off-Grid Mobile Phone Charging," *Energy Procedia*, 2013.

2. Umarani, C., et al., "Solar Docking Station for Smartphones and Laptops," *Materials Today: Proceedings*, 2021.

3. Başoğlu, M. E., et al., "Photovoltaic Mobile Charging System," *Karaelmas Journal of Engineering Sciences*, 2021.

4. Goudar, "Portable Solar-Powered Mobile Charging System," *International Journal for Multidisciplinary Research (IJFMR)*.


# APPENDIX – I

## BILL OF MATERIALS

The Bill of Materials (BOM) presents the components required to construct the proposed Solar Powered Mobile Phone Charger.

**Table A.1 – Bill of Materials**

| S. No. | Component | Quantity | Purpose |
|---|---|---|---|
| 1 | Solar Panel | 1 | Solar energy generation |

| | | | |
|---|---|---|---|
| 2 | Voltage Regulation / DC-DC Module | 1 | Voltage conditioning |
| 3 | USB Output Connector | 1 | Mobile charging interface |
| 4 | Connecting Wires | As required | Electrical connections |
| 5 | USB Cable | 1 | Connection to mobile phone |

# APPENDIX – II

## COMPONENT SPECIFICATIONS AND PROTOTYPE PHOTOGRAPHS

This appendix provides the specifications of the components used in the prototype along with photographs of the completed system.

**Table A.2 – Component Specifications**

| Component | Specification |
|---|---|
| Solar Panel | 5V,1A |
| Voltage Regulation / DC-DC Module | TP4056 Regulator |
| USB Output | USB charging interface |
| Connecting Wires | Suitable for prototype connections |
| USB Cable | Compatible charging cable |

# PROTOTYPE PHOTOGRAPHS

**Figure A.1 – Solar Panel Used in the Prototype**




**Figure A.2 – Voltage Regulation / DC-DC Conversion Module**




**Figure A.3 – Complete Prototype**

